Exercise 3

Let $X_t = AX_{t-1} + \varepsilon_t$ with $p = 2$ and $A = V\Lambda V^{-1}$, for $V = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $\Lambda = \begin{pmatrix} \lambda_1 & 0 \\ 0 & 0.5 \end{pmatrix}$ and $\Omega = \begin{pmatrix} 1 & 3 \\ 3 & 25 \end{pmatrix}$. i) State the stability condition. ii) Fix $\lambda_1 = 0.2$ and with a software of your choice, generate a realization of X_t of length $T = 500$; iii) Estimate A and Ω and compute the characteristic roots; are these close to the values in the DGP? iv) Using the values of A and Ω in the DGP, compute and plot IRFs with the Cholesky decomposition. v) Repeat iv) with the estimated values; are these close to the ones obtained with the values in the DGP? vi) Repeat ii) - v) setting $A_1 = 0.6$. vii) Repeat ii) - v) setting $\lambda_1 = 1$. Repeat ii) – v) setting $\lambda_1 = 1.4$. viii) Comment on how that the results change with respect to λ_1.

Exercise 4

i) Show that D is a decomposition of Ω if and only if $D = D_o Q$, where D_o is the Cholesky decomposition and Q is a rotation matrix. ii) Show that the likelihood function is the same for all D's. iii) Is this is relevant for the analysis? Comment.